

probability $1/2$ and 1 with probability $1/2$. What is the expected running time of your algorithm as a function of p ?

5.2 Indicator random variables

In order to analyze many algorithms, including the hiring problem, we use indicator random variables. Indicator random variables provide a convenient method for converting between probabilities and expectations. Given a sample space S and an event A , the *indicator random variable* $I\{A\}$ associated with event A is defined as

$$I\{A\} = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{if } A \text{ does not occur.} \end{cases} \quad (5.1)$$

As a simple example, let us determine the expected number of heads obtained when flipping a fair coin. The sample space for a single coin flip is $S = \{H, T\}$, with $\Pr\{H\} = \Pr\{T\} = 1/2$. We can then define an indicator random variable X_H , associated with the coin coming up heads, which is the event H . This variable counts the number of heads obtained in this flip, and it is 1 if the coin comes up heads and 0 otherwise. We write

$$\begin{aligned} X_H &= I\{H\} \\ &= \begin{cases} 1 & \text{if } H \text{ occurs,} \\ 0 & \text{if } T \text{ occurs.} \end{cases} \end{aligned}$$

The expected number of heads obtained in one flip of the coin is simply the expected value of our indicator variable X_H :

$$\begin{aligned} E[X_H] &= E[I\{H\}] \\ &= 1 \cdot \Pr\{H\} + 0 \cdot \Pr\{T\} \\ &= 1 \cdot (1/2) + 0 \cdot (1/2) \\ &= 1/2. \end{aligned}$$

Thus the expected number of heads obtained by one flip of a fair coin is $1/2$. As the following lemma shows, the expected value of an indicator

random variable associated with an event A is equal to the probability that A occurs.

Lemma 5.1

Given a sample space S and an event A in the sample space S , let $X_A = I \{A\}$. Then $E[X_A] = \Pr \{A\}$.

Proof By the definition of an indicator random variable from equation (5.1) and the definition of expected value, we have

$$\begin{aligned} E[X_A] &= E[I \{A\}] \\ &= 1 \cdot \Pr \{A\} + 0 \cdot \Pr \{\bar{A}\} \\ &= \Pr \{A\}, \end{aligned}$$

where $\bar{A}$ denotes $S - A$, the complement of A .

■

Although indicator random variables may seem cumbersome for an application such as counting the expected number of heads on a flip of a single coin, they are useful for analyzing situations that perform repeated random trials. In Appendix C, for example, indicator random variables provide a simple way to determine the expected number of heads in n coin flips. One option is to consider separately the probability of obtaining 0 heads, 1 head, 2 heads, etc. to arrive at the result of equation (C.41) on page 1199. Alternatively, we can employ the simpler method proposed in equation (C.42), which uses indicator random variables implicitly. Making this argument more explicit, let X_i be the indicator random variable associated with the event in which the i th flip comes up heads: $X_i = I \{\text{the } i\text{th flip results in the event } H\}$. Let X be the random variable denoting the total number of heads in the n coin flips, so that

$$X = \sum_{i=1}^n X_i.$$

In order to compute the expected number of heads, take the expectation of both sides of the above equation to obtain

$$E[X] = E\left[\sum_{i=1}^n X_i\right]. \quad (5.2)$$

By Lemma 5.1, the expectation of each of the random variables is $E[X_i] = 1/2$ for $i = 1, 2, \dots, n$. Then we can compute the sum of the expectations: $\sum_{i=1}^n E[X_i] = n/2$. But equation (5.2) calls for the expectation of the sum, not the sum of the expectations. How can we resolve this conundrum? Linearity of expectation, equation (C.24) on page 1192, to the rescue: *the expectation of the sum always equals the sum of the expectations*. Linearity of expectation applies even when there is dependence among the random variables. Combining indicator random variables with linearity of expectation gives us a powerful technique to compute expected values when multiple events occur. We now can compute the expected number of heads:

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^n X_i\right] \\ &= \sum_{i=1}^n E[X_i] \\ &= \sum_{i=1}^n 1/2 \\ &= n/2. \end{aligned}$$

Thus, compared with the method used in equation (C.41), indicator random variables greatly simplify the calculation. We use indicator random variables throughout this book.

Analysis of the hiring problem using indicator random variables

Returning to the hiring problem, we now wish to compute the expected number of times that you hire a new office assistant. In order to use a probabilistic analysis, let's assume that the candidates arrive in a random order, as discussed in Section 5.1. (We'll see in Section 5.3 how to remove this assumption.) Let X be the random variable whose value equals the number of times you hire a new office assistant. We could

then apply the definition of expected value from equation (C.23) on page 1192 to obtain

$$E[X] = \sum_{x=1}^n x \Pr\{X = x\} ,$$

but this calculation would be cumbersome. Instead, let's simplify the calculation by using indicator random variables.

To use indicator random variables, instead of computing $E[X]$ by defining just one variable denoting the number of times you hire a new office assistant, think of the process of hiring as repeated random trials and define n variables indicating whether each particular candidate is hired. In particular, let X_i be the indicator random variable associated with the event in which the i th candidate is hired. Thus,

$$\begin{aligned} X_i &= I\{\text{candidate } i \text{ is hired}\} \\ &= \begin{cases} 1 & \text{if candidate } i \text{ is hired ,} \\ 0 & \text{if candidate } i \text{ is not hired ,} \end{cases} \end{aligned}$$

and

$$X = X_1 + X_2 + \cdots + X_n . \tag{5.3}$$

Lemma 5.1 gives

$$E[X_i] = \Pr\{\text{candidate } i \text{ is hired}\} ,$$

and we must therefore compute the probability that lines 5–6 of HIRE-ASSISTANT are executed.

Candidate i is hired, in line 6, exactly when candidate i is better than each of candidates 1 through $i - 1$. Because we have assumed that the candidates arrive in a random order, the first i candidates have appeared in a random order. Any one of these first i candidates is equally likely to be the best qualified so far. Candidate i has a probability of $1/i$ of being better qualified than candidates 1 through $i - 1$ and thus a probability of $1/i$ of being hired. By Lemma 5.1, we conclude that

$$E[X_i] = 1/i . \tag{5.4}$$

Now we can compute $E[X]$:

$$E[X] = E\left[\sum_{i=1}^n X_i\right] \quad (\text{by equation (5.3)}) \quad (5.5)$$

$$= \sum_{i=1}^n E[X_i] \quad (\text{by equation (C.24), linearity of expectation})$$

$$= \sum_{i=1}^n \frac{1}{i} \quad (\text{by equation (5.4)})$$

$$= \ln n + O(1) \quad (\text{by equation (A.9), the harmonic series}) . \quad (5.6)$$

Even though you interview n people, you actually hire only approximately $\ln n$ of them, on average. We summarize this result in the following lemma.

Lemma 5.2

Assuming that the candidates are presented in a random order, algorithm HIRE-ASSISTANT has an average-case total hiring cost of $O(c_h \ln n)$.

Proof The bound follows immediately from our definition of the hiring cost and equation (5.6), which shows that the expected number of hires is approximately $\ln n$. ■

The average-case hiring cost is a significant improvement over the worst-case hiring cost of $O(c_h n)$.

Exercises

5.2-1

In HIRE-ASSISTANT, assuming that the candidates are presented in a random order, what is the probability that you hire exactly one time? What is the probability that you hire exactly n times?

5.2-2

In HIRE-ASSISTANT, assuming that the candidates are presented in a random order, what is the probability that you hire exactly twice?

5.2-3

Use indicator random variables to compute the expected value of the sum of n dice.

5.2-4

This exercise asks you to (partly) verify that linearity of expectation holds even if the random variables are not independent. Consider two 6-sided dice that are rolled independently. What is the expected value of the sum? Now consider the case where the first die is rolled normally and then the second die is set equal to the value shown on the first die. What is the expected value of the sum? Now consider the case where the first die is rolled normally and the second die is set equal to 7 minus the value of the first die. What is the expected value of the sum?

5.2-5

Use indicator random variables to solve the following problem, which is known as the *hat-check problem*. Each of n customers gives a hat to a hat-check person at a restaurant. The hat-check person gives the hats back to the customers in a random order. What is the expected number of customers who get back their own hat?

5.2-6

Let $A[1 : n]$ be an array of n distinct numbers. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an *inversion* of A . (See Problem 2-4 on page 47 for more on inversions.) Suppose that the elements of A form a uniform random permutation of $\langle 1, 2, \dots, n \rangle$. Use indicator random variables to compute the expected number of inversions.

5.3 Randomized algorithms

In the previous section, we showed how knowing a distribution on the inputs can help us to analyze the average-case behavior of an algorithm. What if you do not know the distribution? Then you cannot